Table 1: **Where STRATA sits among related methods.** Keywords describe the mechanism; symbols describe structural properties, not empirical superiority. The final row combines nested subspaces with separate covariance timescales inside the optimizer.

Method / family	Core mechanism	Low-rank moments	Online tracking	Nested subspaces	Multiple covariance EMAs
AdamW (10)	Diagonal adaptivity	✗	✗	✗	✗
GaLore (30)	Periodic SVD	✓	✗	✗	✗
OSD (8)	Online PCA	✓	✓	✗	✗
SubTrack++ (17)	Grassmann tracking	✓	✓	✗	✗
APOLLO (31)	Random-projection scaling	✓	✗	✗	✗
SOAP (25)	Eigenbasis Adam	✗	✗	✗	✗
Muon (6)	Polar update direction	✗	✗	✗	✗
Flag methods (7; 21)	Nested projectors	—	—	✓	—
Single-Grassmann (our control)	One tracked subspace	✓	✓	✗	✗
STRATA (ours)	**Multi-timescale flag state**	✓	✓	✓	✓

✓ present; ✗ absent in the listed formulation; — not an adaptive-optimizer property. *Low-rank moments*: adaptive statistics are stored in compressed coordinates (including APOLLO's auxiliary scaling state). *Online tracking*: incremental data-driven tracking of a compressed gradient subspace, excluding periodic SVD/random refresh and full eigenbasis changes. *Nested*: a nontrivial chain with $K > 1$; a lone subspace is not counted as a multi-level flag. *Multiple covariance EMAs*: distinct decays for the same gradient covariance, not merely Adam's different first-/second-moment decays or SOAP's row/column factors.